

# PE-EC702B — Digital Image and Video Processing

Semester VII • 3L:0T:0P • 3 Credits

## Module 6 Exam Notes: Image Compression

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### Contents

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|          |                                                      |          |
|----------|------------------------------------------------------|----------|
| <b>1</b> | <b>Redundancies in Digital Images</b>                | <b>2</b> |
| <b>2</b> | <b>Lossless Compression Methods</b>                  | <b>2</b> |
| 2.1      | Lossless Predictive Coding                           | 2        |
| 2.2      | Entropy Coding                                       | 2        |
| <b>3</b> | <b>Lossy Compression Methods</b>                     | <b>3</b> |
| 3.1      | Discrete Cosine Transform (DCT) and Transform Coding | 3        |
| <b>4</b> | <b>Still Image Compression Standards</b>             | <b>3</b> |
| 4.1      | JPEG Standard (DCT-Based)                            | 3        |
| 4.2      | JPEG-2000 Standard (DWT-Based)                       | 3        |
| <b>5</b> | <b>Worked Examples</b>                               | <b>4</b> |

## 1. Redundancies in Digital Images

Data compression reduces the amount of data required to represent a given quantity of information by identifying and eliminating statistical redundancies.

### Types of Image Redundancy

- **Coding Redundancy:** Occurs when the codes assigned to gray levels do not minimize average code word length. Resolved by assigning shorter codes to highly probable intensities (entropy coding).
- **Inter-pixel (Spatial) Redundancy:** Results from the high correlation between adjacent pixels. We compress by encoding differences rather than absolute values (predictive coding).
- **Psycho-visual Redundancy:** Information that is ignored by the human visual system (e.g., fine details in highly textured or colored areas). Quantization represents this lossy elimination.

## 2. Lossless Compression Methods

Lossless compression allows perfect reconstruction of the original image.

### 2.1 Lossless Predictive Coding

Based on eliminating inter-pixel redundancy. We predict the current pixel value  $\hat{f}(n)$  based on past pixels, and encode the prediction error (residual):

$$e(n) = f(n) - \text{round} \left[ \sum_{i=1}^m \alpha_i f(n-i) \right] \quad (1)$$

The residual  $e(n)$  has a highly peaked histogram centered at 0, which requires significantly fewer bits when entropy-coded.

### 2.2 Entropy Coding

- **Huffman Coding:** A variable-length prefix coding technique. Assigns shorter binary codes to symbols with high probabilities, and longer codes to low probability ones.
- **Arithmetic Coding:** Maps the entire input symbol sequence into a single floating-point interval  $[0, 1)$ . Highly efficient for skewed probability distributions.
- **LZW (Lempel-Ziv-Welch) Coding:** A dictionary-based algorithm. Replaces repeated sequences of symbols with indices from a dynamically generated dictionary. Used in GIF and TIFF formats.

## 3. Lossy Compression Methods

Lossy compression yields high compression ratios by discarding psycho-visually redundant information, preventing exact image reconstruction.

### 3.1 Discrete Cosine Transform (DCT) and Transform Coding

Transform coding maps pixels into the frequency domain, concentrating energy into a few coefficients. The 2D DCT is defined as:

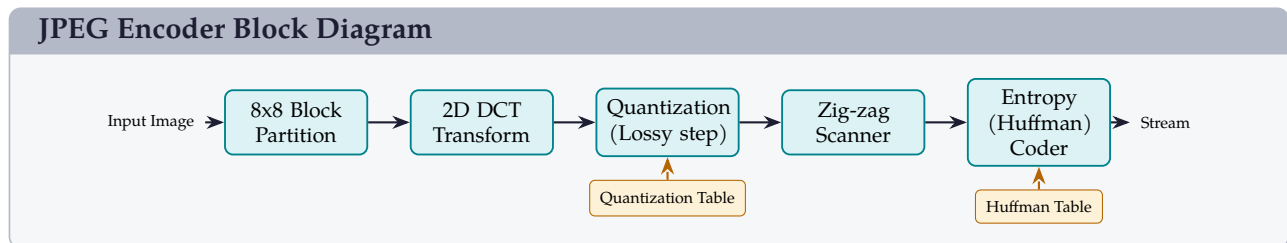
$$T(u, v) = \alpha(u)\alpha(v) \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \cos \left[ \frac{(2x+1)u\pi}{2N} \right] \cos \left[ \frac{(2y+1)v\pi}{2N} \right] \quad (2)$$

where  $N \times N$  is the subimage block size, and  $\alpha(u) = \sqrt{1/N}$  for  $u = 0$ ,  $\alpha(u) = \sqrt{2/N}$  for  $u \neq 0$ .

The DCT has outstanding **energy compaction** (close to KLT) and avoids the blocking boundary discontinuities that DFT introduces.

## 4. Still Image Compression Standards

### 4.1 JPEG Standard (DCT-Based)



1. **Image Partitioning:** Divide the image into non-overlapping  $8 \times 8$  blocks.
2. **DCT:** Apply 2D DCT to each block, producing 1 DC coefficient and 63 AC coefficients.
3. **Quantization:** Divide each coefficient by its corresponding entry in a Quantization Table and round to the nearest integer. *This is the only lossy step in JPEG.*
4. **Zig-zag Scan:** Re-order the 2D matrix into a 1D vector in a zig-zag path, grouping low-frequency components together and producing long runs of zeros.
5. **Entropy Coding:** Compress the values using Run-Length Coding (RLC) and Huffman coding.

### 4.2 JPEG-2000 Standard (DWT-Based)

Instead of DCT, JPEG-2000 uses the **Discrete Wavelet Transform (DWT)**:

- Eliminates the “blocking artifacts” visible in JPEG at high compression ratios.
- Employs EBCOT (Embedded Block Coding with Optimal Truncation).
- Supports progressive transmission (image loads from coarse to fine detail).

## 5. Worked Examples

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## Huffman Coding Design

**Problem:** A discrete source emits 5 symbols with the following probabilities:

| Symbol $s_i$         | $s_1$ | $s_2$ | $s_3$ | $s_4$ | $s_5$ |
|----------------------|-------|-------|-------|-------|-------|
| Probability $P(s_i)$ | 0.4   | 0.2   | 0.2   | 0.1   | 0.1   |

- Construct the binary Huffman code for each symbol.
- Calculate the average code word length  $L_{\text{avg}}$ .
- Compute the source entropy  $H(S)$ .
- Calculate the coding efficiency  $\eta$ .

**Solution:**

**(a) Code Word Construction:** Combine the lowest probabilities recursively:

- Combine  $s_4(0.1)$  and  $s_5(0.1) \rightarrow s_{45}(0.2)$ .
- Sort:  $s_1(0.4)$ ,  $s_2(0.2)$ ,  $s_3(0.2)$ ,  $s_{45}(0.2)$ .
- Combine  $s_3(0.2)$  and  $s_{45}(0.2) \rightarrow s_{345}(0.4)$ .
- Sort:  $s_1(0.4)$ ,  $s_{345}(0.4)$ ,  $s_2(0.2)$ .
- Combine  $s_{345}(0.4)$  and  $s_2(0.2) \rightarrow s_{2345}(0.6)$ .
- Combine  $s_1(0.4)$  and  $s_{2345}(0.6) \rightarrow 1.0$ .

Assigning bits (0 to upper, 1 to lower path):

- $s_1 \rightarrow 0$  (length 1)
- $s_2 \rightarrow 11$  (length 2)
- $s_3 \rightarrow 100$  (length 3)
- $s_4 \rightarrow 1010$  (length 4)
- $s_5 \rightarrow 1011$  (length 4)

**(b) Average Code Word Length ( $L_{\text{avg}}$ ):**

$$L_{\text{avg}} = \sum P(s_i)l_i = 0.4(1) + 0.2(2) + 0.2(3) + 0.1(4) + 0.1(4) = 0.4 + 0.4 + 0.6 + 0.4 + 0.4 = 2.2 \text{ bits/sym}$$

**(c) Entropy ( $H(S)$ ):**

$$\begin{aligned} H(S) &= - \sum P(s_i) \log_2 P(s_i) \\ &= -[0.4 \log_2 0.4 + 2(0.2 \log_2 0.2) + 2(0.1 \log_2 0.1)] \\ &= -[0.4(-1.322) + 0.4(-2.322) + 0.2(-3.322)] \\ &= -[-0.529 - 0.929 - 0.664] = 2.122 \text{ bits/symbol} \end{aligned}$$

**(d) Coding Efficiency ( $\eta$ ):**

$$\eta = \frac{H(S)}{L_{\text{avg}}} \times 100\% = \frac{2.122}{2.2} \times 100\% \approx 96.45\%$$

## Quick Revision — Module VI

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### Key Formulas at a Glance

**Average Code Length**

$$L_{\text{avg}} = \sum_{i=1}^J P(s_i) l_i$$

**Source Entropy**

$$H(S) = - \sum_{i=1}^J P(s_i) \log_2 P(s_i)$$

**Coding Efficiency**

$$\eta = \frac{H(S)}{L_{\text{avg}}} \times 100\%$$

**DPCM Residual Error**

$$e(n) = f(n) - \hat{f}(n)$$

**2D Discrete Cosine Trans.**

$$T(u, v) = \alpha(u) \alpha(v) \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \cos \left[ \frac{(2x+1)u\pi}{2N} \right] \cos \left[ \frac{(2y+1)v\pi}{2N} \right]$$

**Compression Ratio**

$$C_R = \frac{b_{\text{original}}}{b_{\text{compressed}}}$$

### One-Line Definitions

- **Coding Redundancy:** Redundancy occurring when code word lengths do not match the symbols' probability values.
- **Inter-pixel Redundancy:** Spatial correlation between neighboring pixels where values are highly predictable.
- **Psycho-visual Redundancy:** Visual details that can be eliminated without being noticed by human visual perception.
- **DPCM:** A predictive coding technique that encodes only the difference between actual and predicted pixel values.
- **Lossy Compression:** Compression that yields high ratios by discarding psycho-visual data, preventing exact recovery.
- **Zig-zag Scan:** A scan path in JPEG that orders 2D DCT coefficients to group zero values together for run-length coding.

### Frequently Asked / Viva Questions

#### 1. What is the difference between lossless and lossy compression?

Lossless compression reconstructs the original image perfectly without any data loss. Lossy compression discards psycho-visual details to achieve much higher compression, but prevents perfect reconstruction.

#### 2. Which step in JPEG causes loss of information?

The Quantization step. Dividing the DCT coefficients by the quantization table values and rounding them to integers discards high-frequency details.

#### 3. What are blocking artifacts in JPEG?

At high compression ratios, the individual  $8 \times 8$  blocks become visible in the reconstructed image as sharp grid discontinuities due to independent block processing.

**Frequently Asked / Viva Questions (contd.)****4. Why is DCT preferred over DFT for transform coding?**

The DCT has better energy compaction properties and avoids the boundary discontinuities that the DFT introduces, reducing blocking artifacts.

**5. How does JPEG-2000 eliminate blocking artifacts?**

It uses the Discrete Wavelet Transform (DWT) on the entire image (or large tiles) rather than the DCT on small  $8 \times 8$  blocks.

**6. What is prefix code in Huffman coding?**

A prefix code is a code in which no code word is a prefix of any other code word, ensuring that the code can be uniquely decoded in real-time.

**7. Define Shannon's First Source Coding Theorem.**

It states that the minimum average code length required to represent a source without distortion is bounded below by the source entropy  $H(S)$ .

**8. Explain LZW coding.**

It is a dictionary-based, lossless compression method that replaces recurring sequences of symbols with dictionary indices, commonly used in GIF files.

**9. What is the function of the zig-zag scan in JPEG?**

It converts the 2D block of quantized coefficients into a 1D sequence, grouping high-frequency zero coefficients together to optimize Run-Length Encoding.

**10. How does Arithmetic Coding differ from Huffman Coding?**

Huffman coding maps each individual symbol to a bit sequence. Arithmetic coding encodes the entire message as a single fractional value within an interval.

**JPEG vs. JPEG-2000 Comparison**

| Attribute             | JPEG                                   | JPEG-2000                              |
|-----------------------|----------------------------------------|----------------------------------------|
| Transform Method      | Block-based 2D DCT ( $8 \times 8$ )    | Full-image 2D DWT (Wavelet)            |
| Compression artifacts | Blocking artifacts at high compression | Blurring / ringing at high compression |
| Progressive Display   | Poor support                           | High support (resolution scalability)  |